

- ① Which of the following formulas is used to compute lower triangular matrix entries in Crout's algorithm?

$$l_i = a_i - \frac{c_{i-1}}{l_{i-1}} b_i, \text{ for } i=1, 2, 3, \dots, n$$

- ② In Crout's method of LU factorization matrix, U has 1's on its diagonal.

- ⑤ Which inequality is sufficient condition for convergence of Jacobi's iterative method?

$$|a_{ii}| \geq \sum_{j \neq i}^n |a_{ij}|, \quad i=1, 2, \dots, n$$

- ④ The Schmidt method applied on 1D heat equation is a two level scheme.

- ③ The Crank-Nicolson method is unconditionally stable for unsteady diffusion equation.

- ⑥ Another name of Jacobi's iterative method is displacement method.

- ⑦ The process of constructing a sequence and obtaining the solution of a system using specified accuracy is called iteration.

- ⑧ Which PDE generates First Order upwind scheme after applying appropriate difference approximations and $c > 0$.

$$\frac{\partial f(x,t)}{\partial t} + c \frac{\partial f(x,t)}{\partial x} = 0$$

- ⑨ A finite difference method when applied on PDE will be stable if amplification factor G is bounded.

- ⑩ The Crank-Nicolson scheme when applied on heat diffusion equation is unconditionally stable.

- ⑪ Laplace equation is a type of elliptical partial eq.

- ⑫ Which equation represents 2D Laplace PDE?
- $$\nabla^2 u = 0$$

- ⑬ The discretized form of 2D Laplace PDE for uniform grid spacing in both direction is given by:
- $$4w_{i,j} = [(w_{i-1,j} + w_{i+1,j}) + (w_{i,j-1} + w_{i,j+1})]$$

- ⑭ Consider the mesh point $w_I = w_{i,j}$ of solution domain, where I denotes the labeling to internal grid points and I is defined as $I = j + (m-1-i)(n-1)$

- ⑮ The discretized form of 2D Poisson PDE when central difference approximation is applied on both derivatives with uniform step size $h=0.5$ in

both direction is,

$$4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = -f_{i,j}$$

- (16) The discretized form of 2D Poisson PDE when central difference approximation is applied on both derivative with uniform step size $h=0.5$ in both directions is

$$4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = \frac{1}{4} f_{i,j}$$

- (17) The solution of elliptical eq. depends on all of its boundaries

- (18) Which is an elliptic PDE? Laplace equation

- (19) Gauss-Seidel method is similar to which of method.
Jacobi's method

- (20) Gauss-Seidel method requires more iterations than Jacobi's method. False

- (21) Which step is not involved in factorization process?
elimination of unknown using back substitution

- (22) Which is name of factorization method?
Doolittle's method

- (23) The diagonal 5-point formula when applied on Laplace eq. remains invariant on coordinate axes.

(24) The acceleration factor ω in SOR method lies between **[1, 2]**

(25) In the eq. below, when SOR method is applied on 2D Laplace eq. $u_{i,j}^{(n+1)} = u_{i,j}^{(n)} + \frac{\omega}{4} R_{i,j}$, the parameter ω represents the **acceleration factor**.

(26) Gauss-Seidel method is a special case of SOR method if acceleration method $\omega = 1$.

(27) The SOR method is more powerful iterative method than **both Jacobi & Gauss-Seidel method**.

(28) Alternating Direction Implicit (ADI) method is **iterative method**.

(29) The ADI method represented by the equation $u_{i-1,j}^{(n+1)} - 4u_{i,j}^{(n+1)} + u_{i+1,j}^{(n+1)} = -u_{i,j-1}^{(n)} - u_{i,j+1}^{(n)}$ is used to compute the values at all internal mesh points **along the rows**.

(30) The diagonal 5-point formula is obtained when coordinate axis is rotated through an angle of **45** degree.

(31) The discretized PDE $u_i^{(n+1)} = 2(1-r^2c^2)u_i^n + r^2c^2(u_{i-1}^n + u_{i+1}^n) - u_i^n$ is a **three** level scheme.

(32) Telegraph PDE is a type of hyperbolic PDE.

(33) While solving a Hyperbolic PDE, the first-level solution can be obtained from boundary condition.

(34) Damped wave equation is hyperbolic PDE.

(35) Which is not a type of flows based on their mathematical behaviour? circular

(36) The PDE is classified as $5 \frac{\partial^2 u}{\partial x^2} + 6 \frac{\partial^2 u}{\partial y^2} = xy$
elliptic.

(37) The method of lines (MOL) is generally used for space dependent PDE.

(38) In MOL, finite difference approximations for special derivatives are used to convert a PDE to a large set of ordinary differential equation.

(39) The region in which PDE
$$x^3 \frac{\partial^2 u}{\partial x^2} + 3 \frac{\partial^2 u}{\partial x \partial y} + 27 \frac{\partial^2 u}{\partial y^2} = 0$$

acts as an elliptic equation is

(40) For the PDE

$$3 \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + 3 \frac{\partial^2 u}{\partial y^2} = 0$$

to be classified as Elliptic, the value of B must be

Solution

Since $|a_{ii}| \geq \sum_{j=1}^n |a_{ij}|$

$$|a_{11}| \geq |a_{12}| + |a_{13}|$$

$$|3| \geq |-2| + |1| \Rightarrow |3| \geq 2 + 1$$

$$\Rightarrow 3 = 3$$

$$|a_{22}| \geq |a_{21}| + |a_{23}|$$

$$|-3| \geq |1| + |2| \Rightarrow 3 = 1 + 2$$

$$3 = 3$$

$$|a_{33}| \geq |a_{31}| + |a_{32}|$$

$$|4| \geq |2| + |2| \Rightarrow 4 > 3$$

So, the matrix A is diagonally dominant.

Qno. 44

Sol

$$l = i + (m-1-j)(n-1)$$

As $w_{i,j} = w_l$ and $n=m=10$

at $w_{2,4} \Rightarrow i=2, j=4$

So

$$l = 2 + (10-1-4)(10-1) = 47$$

$$w_{2,4} = w_{47}$$

at $w_{3,5} \Rightarrow i=3, j=5$

So

$$l = 3 + (10-1-5)(10-1) = 39$$

$$w_{3,5} = w_{39}$$

Q no. 49

Solution:-

Applying CD approximation on 2D Laplace equation we get

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{(\Delta x)^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{(\Delta y)^2} = 0$$

let $\Delta x = \Delta y = h$

So above equation become

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2} = 0$$

After Solving above equation we get

$$u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$$

Solution

Jacobi's SOR scheme for Poisson equation using rotated five points formula is given as

$$u_{i,j}^{(n+1)} = \frac{1}{4}w \left[u_{i-1,j-1}^{(n)} + u_{i+1,j-1}^{(n)} + u_{i,j-1}^{(n)} + u_{i,j+1}^{(n)} + h^2 f_{i,j} \right] + (1-w) u_{i,j}^{(n)}$$

Gauss-Seidel SOR Iterative scheme for Poisson equation using rotated five points formula is

$$u_{i,j}^{(n+1)} = \frac{1}{4}w \left[u_{i-1,j-1}^{(n+1)} + u_{i+1,j-1}^{(n+1)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)} + h^2 f_{i,j} \right] + (1-w) u_{i,j}^{(n)}$$

Solution:-

Jacobi's OR method for Poisson
eq can be written as

$$u_{i,j}^{(n+1)} = \frac{1}{4} \omega \left[u_{i-1,j}^{(n)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n)} + u_{i,j+1}^{(n)} + h^2 f_{i,j} \right] + (1-\omega) u_{i,j}^{(n)}$$

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Q no. 41

$$T_i^{n+1} = \frac{2\lambda}{1+2\lambda} [T_{i-1}^n + T_{i+1}^n] + \frac{1-2\lambda}{1+2\lambda} T_i^{n-1}$$

Qno. 47

Solution

$$l_1 = a_1 \Rightarrow l_1 = 4$$

$$l_2 = a_2 - \frac{C_1}{\lambda_1} b_2$$

$$l_2 = 4 - \frac{-1}{4} (-1) \Rightarrow l_2 = \frac{15}{4} = 3.75$$

$$l_3 = a_3 - \frac{C_2}{l_2} b_3$$

$$l_3 = \frac{56}{15} = 3.73$$

$$l_4 = 3.73$$

$$l_5 = 3.73$$